DextraVerse: A Unified Platform for Tactile-Rich Bimanual Dexterous Manipulation

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***Abstract*— Diverse hardware, sensing modalities, and learning algorithms offer complementary capabilities for dexterous manipulation, but fragmented implementations hinder their systematic investigation and integration. We present DextraVerse, a unified and versatile research platform that addresses this fragmentation through shared control and data interfaces connecting teleoperation, multimodal sensing, imitation learning (IL), reinforcement learning (RL), and execution in simulation and on real robots. These interfaces support two complementary capabilities: identifying common patterns across learning paradigms and combining platform components into new manipulation workflows. Tactile representation benchmarks across IL and RL reveal that compact force-based inputs can improve task performance, while deformation maps offer substantial benefits in selected evaluations but remain challenging to exploit consistently. To demonstrate component integration, we use simulation-trained RL hand skills to assist real-world demonstration collection for downstream IL, enabling a single operator to collect approximately 75 demonstrations per hour. Tactile-informed rewards provide another example of component integration, using contact information to guide finger participation and grasp configuration. To support further extension of these workflows, DextraVerse provides Repository Retrieval Maps for AI-assisted development. A seven-agent study reports average reductions of 51.3% in repository-inspection calls and 25.2% in token usage with these maps across the full sample. Together, these studies demonstrate DextraVerse's utility as a reusable experimental foundation for investigating sensing and learning choices and integrating complementary components for multifingered dexterous manipulation.**

I. Introduction

Dexterous manipulation research spans diverse robot embodiments, teleoperation interfaces, sensing modalities, and learning algorithms. However, these advances are commonly developed within separate hardware stacks, simulation environments, data pipelines, and deployment systems, making it difficult to compare alternatives and integrate them into manipulation workflows. This necessitates a unified experimental platform that enables researchers to identify insights that remain consistent across learning paradigms, distinguish them from task-specific effects, and systematically integrate complementary approaches for dexterous manipulation.

At the policy-learning level, imitation learning (IL) and reinforcement learning (RL) illustrate both the opportunities and challenges of integrating complementary methods. IL learns complex visuomotor behaviors directly from demonstrations across diverse real-world manipulation tasks [1], [2], but its performance depends on the coverage and quality of training data. RL, on the other hand, learns through interaction and has demonstrated capabilities in high-DoF, contact-rich control [3], [4], [5], [6], [7]. RL can acquire skills without expert demonstrations, while simulation-based

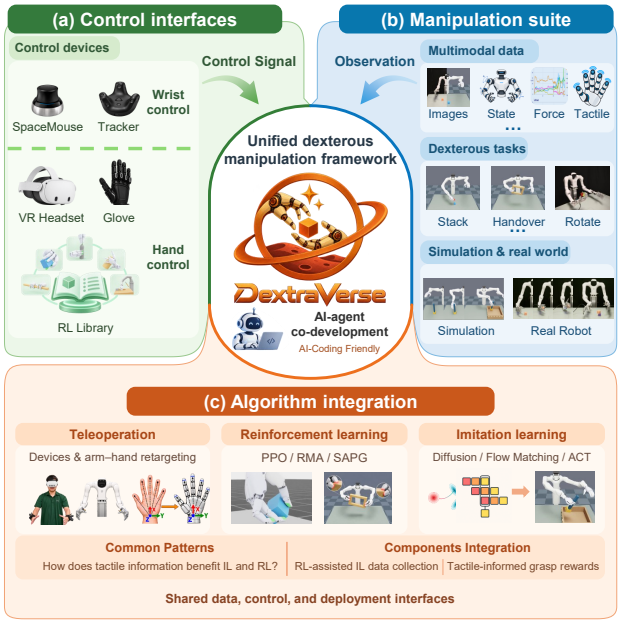


Fig. 1: **Overview of DextraVerse.** DextraVerse unifies (a) control interfaces, (b) multimodal manipulation in simulation and the real world, and (c) teleoperation, IL, and RL through shared interfaces, with support for AI-agent co-development.

RL requires careful reward design and sim-to-real transfer for real-world deployment. Moreover, its advantages over conventional control remain task-dependent [8]. These complementary strengths motivate both controlled comparison and method integration, such as using simulation-trained RL hand skills to assist teleoperation and collect demonstrations for downstream IL. Supporting these workflows calls for shared control and data interfaces across policy training and deployment in both simulation and the real world.

"Harmony in Diversity, Innovation via Integration."

Guided by this principle, we introduce **DextraVerse**, a unified platform for bimanual dexterous manipulation built upon a 58-DoF robotic embodiment. As illustrated in Fig. 1, *"Harmony in Diversity"* is realized through the compatible, modular wrist and hand interfaces in (a), which support a manipulation suite spanning simulation and real-world hardware in (b). *"Innovation via Integration"* is enabled by the shared observation-action and data interfaces in (c), allowing researchers to integrate heterogeneous components across manipulation workflows. Structured repository documentation and experiment tooling further support the extension of these capabilities, providing a foundation for continued research and future advances in robotics.

We demonstrate DextraVerse's versatility through two complementary capabilities, illustrated in Fig. 1(c): identifying **common patterns across learning paradigms** and enabling **component integration**. First, shared data and observation–action interfaces support tactile representation benchmarking across IL and RL for multifingered dexterous manipulation. Comparisons of three tactile information types and their combinations show that appropriate tactile inputs can benefit both paradigms, while richer observations do not consistently improve performance. Second, DextraVerse enables researchers to combine platform components into new manipulation workflows. Simulation-trained RL hand skills assist real-world teleoperation and demonstration collection for downstream IL, achieving approximately 75 demonstrations per hour with a single operator. Tactile-informed rewards provide another example, using contact information to guide finger participation and grasp gesture. Repository Retrieval Maps further support platform development by helping AI coding agents locate implementations and trace dependencies. Our main contributions are as follows:

- We present **DextraVerse**, a modular research platform unifying teleoperation, multimodal sensing, IL/RL, and execution in simulation and on real robots through shared interfaces for tactile-rich dexterous manipulation.
- DextraVerse enables **tactile representation benchmarking** across IL and RL, supporting comparisons of force magnitude, force direction, deformation maps, and their combinations across dexterous manipulation tasks.
- DextraVerse allows **component composition**: simulation-trained RL hand skills assist real-world teleoperation and demonstration collection for IL, while tactile-informed rewards guide finger participation and grasp configuration.
- DextraVerse provides **Repository Retrieval Maps** to help AI coding agents navigate and extend the integrated robotic system, with a seven-agent study demonstrating reduced repository-inspection calls and token usage.

We will open-source DextraVerse to provide a reusable infrastructure for future research in dexterous manipulation.

II. Related Work

A. Teleoperation and Manipulation Platforms

Robotic manipulation platforms integrate hardware, sensing, control, and learning components to support experimental development and evaluation. Teleoperation tools enable human-guided control and demonstration collection: DexPilot [9] and AnyTeleop [10] support vision-based dexterous teleoperation, while DexCap [11] enables portable demonstration collection. At the platform level, robosuite [12] provides simulation and learning infrastructure, while DexVerse [13] and DexJoCo [14] integrate task suites, data collection, and policy evaluation. Table I compares the documented capabilities of representative platforms. Within this comparison, DextraVerse brings together bimanual manipulation, simulation and real-world execution, tactile sensing, and support for both IL and RL.

TABLE I: Platform capabilities. All support simulation; "–" means not documented in the cited work.

Platform	Year	Bimanual	Real	Tactile	IL	RL
robosuite [12]	2020	✓	–	–	✓	✓
DexJoCo [14]	2026	✓	–	–	✓	–
DexVerse [13]	2026	✓	–	–	✓	–
TeleDexter [18]	2026	–	✓	–	✓	✓
DextraVerse	Ours	✓	✓	✓	✓	✓

B. Imitation Learning and Reinforcement Learning

IL and RL provide complementary approaches to learning manipulation skills: IL learns from demonstrations, while RL learns through interaction under task-specific rewards. ACT [2], Diffusion Policy [1], and vision–language–action (VLA) models [15], [16] demonstrate the capabilities of demonstration-based learning for manipulation. For dexterous hand control, SimToolReal [17] enables real-world tool use through policies trained in simulation. IL and RL can also be composed across learning stages: TeleDexter [18] uses simulation-trained RL controllers to assist dexterous teleoperation and collect demonstrations for behavior cloning. Such workflows motivate reusable interfaces that connect demonstration collection, policy learning, and execution across tasks and control configurations.

C. Tactile Sensing for Dexterous Manipulation

Tactile sensing provides contact information for precise, contact-rich dexterous manipulation. Binary contact signals enable vision-free in-hand rotation [19], while ViTacFormer [20] and Reactive Diffusion Policy [21] investigate multimodal fusion and tactile feedback. VTDexManip [22] studies visual-tactile pretraining for dexterous RL. These studies motivate systematic investigation of how tactile representation choices influence learning and whether their benefits extend across tasks and learning methods.

Investigating these questions and composing IL–RL workflows require infrastructure that supports controlled comparisons and component reuse. DextraVerse provides shared data structures and observation–action interfaces that enable systematic comparisons to identify common patterns across tactile-rich dexterous manipulation tasks and learning methods. These interfaces also enable researchers to integrate complementary sensing, control, and learning components into unified workflows for bimanual dexterous manipulation.

III. Methodology

A. Platform Design

DextraVerse is a unified, AI-coding-friendly platform with three layers (Fig. 2). The *control interface* generates commands through teleoperation or policy inference; *robot execution* applies commands in simulation or on hardware and returns multimodal observations; and *policy learning* trains IL policies and RL skills for control. Shared data structures and control interfaces support the combination and reuse of sensing, control, and learning components across layers.

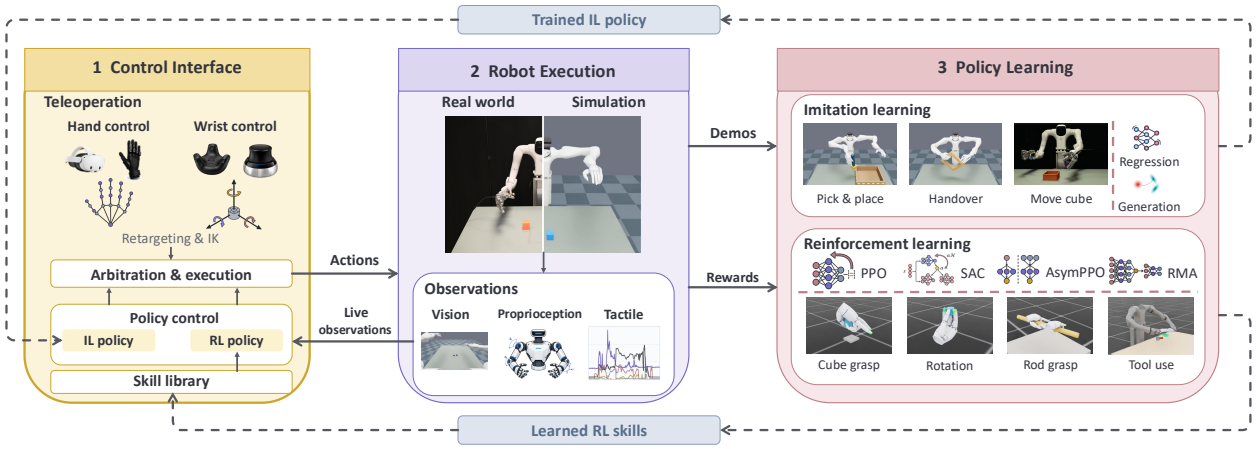


Fig. 2: **Overview of DextraVerse's three-stage workflow: control interface, robot execution, and policy learning.** Shared interfaces connect teleoperation and policy control with real and simulated robots, supporting multimodal data collection, imitation learning, and reinforcement learning. Trained IL policies and learned RL skills feed back into the control interface for deployment.

Repository Retrieval Maps guide AI coding agents to relevant code and interfaces, supporting platform extension.

B. Hardware and Demonstration Collection

1) Hardware: We instantiate DextraVerse on the Dexmate Vega platform with two 7-DoF robotic arms and two 22-DoF Sharpa dexterous hands equipped with fingertip visual-tactile sensing. A corresponding simulation model in Isaac Lab preserves the robot geometry and kinematics and provides task-specific sensing and control interfaces for simulation-based learning. Simulated contact signals approximate physical sensing, while calibration and command conversion account for sensing and control differences during real-world deployment. Teleoperation is decomposed into wrist and hand control. Six-DoF wrist targets are derived from Vive trackers with base stations or incremental Cartesian commands from a 3D SpaceMouse, then converted into arm joint targets using inverse kinematics (IK). Human hand skeletons obtained from VR hand tracking or data gloves are retargeted to robotic finger joint targets. This modular decomposition allows different wrist and hand interfaces to be combined within a common robot-control pipeline.

2) Demonstration Collection: DextraVerse records robot observations and commanded actions through a common interface across supported teleoperation devices. Recorded observations include multi-view RGB images, arm and hand proprioception, end-effector poses, and configured fingertip tactile inputs: force magnitude, force direction, and deformation maps. Actions are recorded as joint targets under the shared control conventions used for teleoperation and IL/RL policy execution. Data validity is checked during collection, and compatibility of embodiment, action, and sensing configurations is verified before demonstrations are merged for downstream policy learning. The same recording interface supports capturing human corrections during policy deployment. Because arm and hand control can be assigned to separate pipelines, an operator can correct arm motion while an RL controller continues to control the hand. Corrected trajectories are stored alongside the corresponding policy actions and correction metadata for subsequent analysis.

C. Unified Policy Learning Layer

DextraVerse provides shared data structures and observation–action interfaces to support the integration of IL and RL. Within each learning pipeline, researchers can vary sensing modalities and learning algorithms while reusing task, data, and evaluation infrastructure. Across pipelines, learned controllers can be combined with other platform components to connect policy learning, teleoperation, and robot execution.

1) Imitation Learning: DextraVerse supports learning observation-conditioned action policies from demonstrations collected through the shared recording interface. The IL framework combines a manipulation task suite with configurable policy algorithms and sensory inputs, enabling systematic comparisons across different dexterity requirements. ***Manipulation Tasks:*** The simulated task suite includes *Pick-and-place*, *Handover*, and *Cube stacking*, as shown in Fig. 3(a-c), which assess complementary capabilities. *Pick-and-place* requires grasping a hammer and placing it into a basket, assessing grasp acquisition and maintenance. *Handover* requires transferring a wood frame between two hands, emphasizing bimanual coordination and contact stability during transfer. *Cube stacking* requires placing a red cube onto a blue cube, highlighting grasp control, object alignment, and placement precision. The real-world cube-to-basket task (Fig. 3(d)) extends the workflow to physical demonstration collection, IL training, and policy deployment. ***Policy Algorithms:*** The IL framework supports Diffusion Policy (DP) [1], Flow Matching (FM) policies [23], Action Chunking with Transformers (ACT) [2], MeanFlow-based policies [24], and Action-to-Action flow matching (A2A) [25]. Shared data and evaluation interfaces support algorithm comparisons within each task, while configurable sensory inputs enable representation studies within a selected algorithm. Trained policies use the shared control interface for evaluation in simulation and deployment on the robot.

2) Reinforcement Learning: DextraVerse supports simulation-based dexterous skill learning through configurable tactile observations, modular reward definitions, and a shared skill library. These components enable researchers to investigate how tactile inputs and reward design affect learned behaviors and reuse trained controllers in teleoperation and IL data collection.

Reward Design: DextraVerse provides modular reward definitions that support the construction and comparison of manipulation objectives. Each reward term specifies three attributes: (i) a *semantic identity* describing the behavior or condition being rewarded; (ii) a *reward role*, categorized as shaping, terminal, or penalty; and (iii) a reusable *mathematical kernel* defining its computation. This separation allows reward terms to reuse computational kernels while retaining distinct behavioral meanings. Researchers can compose objectives from existing terms or modify selected terms while reusing the training and evaluation infrastructure, supporting systematic investigation of reward design.

Dexterous Skill Library: DextraVerse provides a shared skill library covering grasping, in-hand manipulation, and bimanual manipulation. Each skill specification connects a reward objective with its training protocol and evaluation settings, including curricula, randomization, seeds, and task metrics. This organization supports the addition and assessment of skills within the common RL framework. The shared control interface makes learned controllers available to other platform components, subject to compatible observation and action requirements. Separate arm and hand control allows learned skills to be combined with other control pipelines, supporting workflows that connect simulation training, teleoperation, and downstream policy learning.

D. AI-Coding-Friendly Design

DextraVerse facilitates AI-assisted platform development through *Repository Retrieval Maps (RRMs)*, interface definitions, and automated checks. RRMs guide coding agents from a global package index to subsystem maps identifying relevant implementations, interfaces, and tests, helping them trace dependencies across sensing, control, and learning components. Interface definitions specify observation shapes, coordinate frames, and joint orderings to clarify compatibility requirements when connecting or modifying components.

Machine-readable configurations, automated checks, and run metadata support validation and experiment reproduction. The seven-agent evaluation in Sec. IV-D compares the same code with and without RRMs. Across the full sample, RRMs reduce inspection calls, token usage, and wall-clock time by averages of 51.3%, 25.2%, and 35.4%, respectively.

IV. Experiments

We demonstrate the versatility of DextraVerse through systematic investigation and component integration. Tactile representation studies across simulated IL and RL examine common patterns and task-dependent effects, while comparisons of IL algorithms demonstrate policy evaluation within the shared task suite. Component integration is demonstrated through RL-assisted real-world demonstration collection for IL and tactile-informed reward design for shaping grasp configuration. Finally, a seven-agent study evaluates how Repository Retrieval Maps improve repository-navigation efficiency and support AI-assisted platform development.

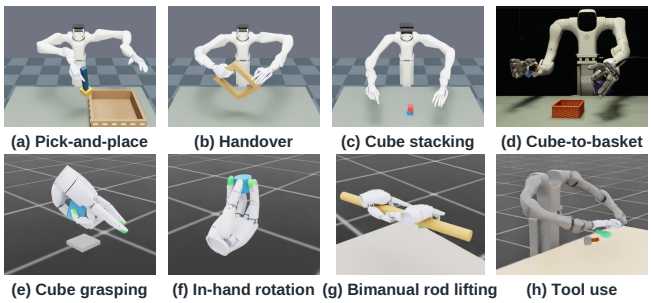


Fig. 3: **Manipulation tasks in DextraVerse.** (a–c) simulated IL benchmarks; (d) real-world demonstration collection and IL deployment; (e–h) simulated RL benchmarks.

A. Tactile Representation Benchmarking

Understanding how tactile information benefits dexterous manipulation requires examining its effects across learning paradigms and tasks. DextraVerse enables this investigation through shared learning interfaces and configurable tactile observations. We use this capability to identify common patterns across IL and RL and examine how the benefits of tactile information vary across tasks.

Task Suite: Simulated IL tasks include *Pick-and-place*, *Handover*, and *Cube stacking* (Fig. 3, a–c), with 30 demonstrations per task collected through teleoperation using VR hand pose detection and Vive Tracker wrist tracking. Simulated RL tasks include *Cube grasping*, *In-hand rotation*, *Bimanual rod lifting*, and *Tool use* (Fig. 3, e–h). We compare six tactile-input configurations: T0, no tactile input; T1, force magnitude; T2, force magnitude and direction; T3, deformation maps; T4, deformation maps and force magnitude; and T5, all three types. Settings are matched within each study; cross-paradigm analyses examine common qualitative patterns and differences across tasks under task-specific protocols.

Tactile Inputs: For each fingertip i, force magnitude, force direction, and deformation maps are represented as $\mathbf{m}^{(i)} \in \mathbb{R}^{T\times 1}$, $\mathbf{d}^{(i)} \in \mathbb{R}^{T\times 3}$, and $\mathbf{M}^{(i)} \in \mathbb{R}^{T\times 24\times 24}$, respectively, where T denotes the sequence length. These inputs describe contact strength, force orientation, and spatial deformation. Simulated deformation maps are contact-geometry-based proxies for fingertip surface deformation.

1) Tactile Ablation in IL: We compare T0–T5 using a flow-matching policy [23], training each configuration for 1,000 epochs. The left side of Table II shows that compact tactile inputs can benefit IL. Force magnitude with direction (T2) exceeds the tactile-free baseline in mean success on all three tasks, whereas force magnitude alone (T1) only improves pick-and-place and cube stacking. T1 and T2 tie for the best mean rank (2.67), although some improvements are small relative to the reported variability.

Deformation-map configurations show potential but provide less consistent benefits. Maps alone (T3) achieve the

TABLE II: Tactile-representation ablation across simulated IL and RL manipulation tasks with seeds (42, 123, 456).

Rep.	IL: task success (%) ↑			Rank ↓	RL: task performance (%) ↑			Rank ↓
	Pick-and-place	Handover	Cube stacking		Cube grasping	In-hand rotation	Bimanual rod lifting	
T0	36.7 ± 3.3	16.7 ± 15.3	1.1 ± 1.9	4.00	71.7 ± 11.5	89.5 ± 1.8	40.5 ± 38.5	4.67
T1	47.8 ± 1.9	11.1 ± 10.2	3.3 ± 0.0	**2.67**	91.1 ± 1.3	89.4 ± 3.3	23.2 ± 34.8	4.00
T2	42.2 ± 18.4	**18.9** ± 6.9	2.2 ± 3.9	**2.67**	**92.9** ± 0.7	93.6 ± 0.8	68.4 ± 16.4	**1.67**
T3	**52.2** ± 12.6	7.8 ± 7.7	0.0 ± 0.0	4.33	88.5 ± 2.8	88.2 ± 4.0	45.7 ± 38.7	4.33
T4	40.0 ± 11.6	11.1 ± 10.7	3.3 ± 0.0	3.33	86.6 ± 7.8	84.5 ± 7.6	66.3 ± 23.9	4.67
T5	17.8 ± 6.9	10.0 ± 10.0	**11.1** ± 3.9	4.00	91.1 ± 1.1	**93.8** ± 4.5	**77.7** ± 4.0	**1.67**

highest pick-and-place success, increasing it from 36.7% to 52.2%, while maps combined with force magnitude and direction (T5) achieve the highest stacking success, increasing it from 1.1% to 11.1%. However, T3 falls below the baseline on handover and stacking, and T5 falls below it on pick-and-place and handover. Our current implementation encodes deformation maps using a ResNet and concatenates the resulting features with other inputs to condition the policy. These mixed results motivate further investigation into encoding and fusion methods that can exploit spatial tactile information more consistently.

2) Tactile Ablation in RL: To identify which types of tactile representation might benefit the RL skill training, we evaluate the six types of tactile representations T0–T5 on three tasks, *Cube grasping*, *In-hand rotation*, and *Bimanual rod lifting*, and then also provide the evaluation on the multi-stage task *Tool use* [17] with lift and goal reaching.

Evaluation Protocol: We compare T0–T5 under the training configurations in Table III, holding task settings, rewards, and curricula fixed within each task while allowing input dimensions and encoder capacity to vary. Each policy is evaluated using 256 environments and 600 rollout steps with task-specific metrics. For *Cube grasping* and *Bimanual rod lifting*, we report the fraction of environments and rollout steps satisfying the respective hold or lift criterion: cube hold requires reaching 90% of the target height while remaining below the speed threshold, whereas rod lift follows the bimanual-lift criterion. *In-hand rotation* measures episode success, requiring at least 10 rad of net rotation without dropping the object. For tool use, we report both object lift success rate and goals reaching achieved.

To distinguish intermediate performance from final outcomes, we evaluate the reward-best and final checkpoints. For single-stage tasks, we report the higher evaluated task score per run; for *Tool use*, we report both checkpoints separately and additionally assess long-horizon training stability. We quantify stability through reward retention, defined as the mean smoothed return over the final 200 million steps divided by its peak, with retention below 0.35 indicating collapse. To locate the onset of collapse, we identify the first post-peak decrease below half the peak without subsequent recovery above that threshold.

Single-Stage Manipulation: As shown in Table II, force magnitude and direction (T2) exceed the tactile-free baseline in the reported mean task metric on all three single-stage tasks. T2 achieves the highest cube hold fraction and ties T5 for the best mean rank, computed from unrounded task means. This agrees with the IL observation that compact force-based inputs can benefit policy learning. Combining these force cues with deformation maps (T5) yields the highest rod lift fraction and the smallest observed range across runs (72.8–80.8%). T2, T4, and T5 achieve nonzero rod lift fractions in every run, whereas T0, T1, and T3 exhibit substantially greater variability. However, deformation maps alone (T3) do not consistently outperform force-based inputs.

TABLE III: Training configurations for RL tactile ablations.

Task	Algorithm	Envs.	Steps
Cube grasping	PPO [26]	8,192	300M
In-hand rotation	PPO [26]	8,192	300M
Rod lifting	Asymmetric PPO [27]	4,096	300M
Tool use	Asymmetric PPO [27]	6,144	5B

Fig. 4 complements these task outcomes with training returns. Several tactile configurations show faster reward growth than the tactile-free baseline, but T3 and T4 exhibit substantial variability on bimanual rod lifting. Richer tactile observations therefore do not ensure consistent learning.

These mixed benefits also warrant consideration of computational cost. RTX 5090 probes estimate 2.06–2.50 GPU-hours per 300 million agent steps across the single-stage tasks. Relative to T0, force-based inputs change peak VRAM usage by at most approximately 1%, whereas map-based inputs increase it by 13–33%. T2 thus offers a favorable combination of task performance and memory usage in these experiments, while deformation maps incur greater memory costs without consistently improving performance.

Multi-Stage Tool Use: The tool-use task comprises two stages: object lifting followed by goal reaching. *Lift Rate* measures the first stage, while *Goals Reaching* reports the number of consecutive goals achieved in the second stage. As shown in Table IV, all three configurations incorporating deformation maps (T3–T5) exceed the tactile-free baseline on goal reaching at the reward-best checkpoint. T5 achieves the highest lift rate (77.0%) and goal-reaching score (2.50, compared with 0.50 for T0) at this checkpoint.

The gains from deformation maps are not uniformly sustained. Goal-reaching scores decrease between the reward-best and final checkpoints for T3–T5, with T4 and T5 falling below the final tactile-free baseline. T5 shows the largest decline: its lift rate decreases from 77.0% to 55.0%, while its goal-reaching score drops from 2.50 to 0.48. T3's goal-reaching score declines more modestly, from 1.09 to 0.94, and remains the highest at the final checkpoint.

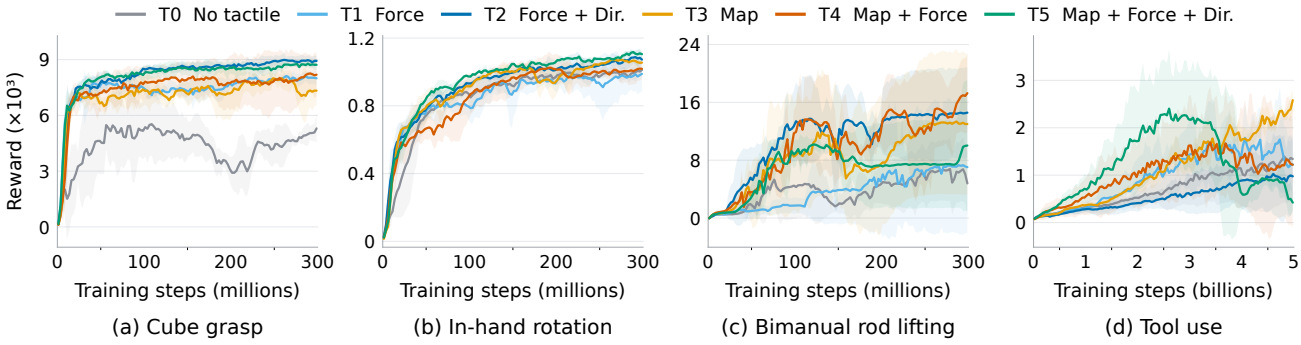


Fig. 4: **Training rewards** for (a) cube grasping, (b) in-hand rotation, (c) bimanual rod lifting, and (d) tool use under three training seeds. Colors denote T0–T5 as the combinations of Force Magnitude (Force), Force Direction (Dir.), and Deformation Map (Map).

TABLE IV: Long-horizon tactile ablation on *Tool use*. Values are mean ± standard deviation across evaluation seeds.

Rep.	Lift Rate(%) ↑		Goal Reaching ↑	
	Reward-best	Final	Reward-best	Final
T0	70.0 ± 2.0	69.0 ± 1.0	0.50 ± 0.03	0.61 ± 0.05
T1	67.0 ± 1.0	54.0 ± 1.0	0.60 ± 0.03	0.41 ± 0.03
T2	73.0 ± 2.0	**71.0** ± 1.0	0.22 ± 0.04	0.29 ± 0.04
T3	71.0 ± 0.0	69.0 ± 1.0	1.09 ± 0.08	**0.94** ± 0.02
T4	72.0 ± 1.0	66.0 ± 3.0	0.95 ± 0.02	0.46 ± 0.01
T5	**77.0** ± 1.0	55.0 ± 1.0	**2.50** ± 0.05	0.48 ± 0.02

Training traces further distinguish these outcomes. Two of three T5 runs collapse near 3.58 and 3.95×10^9 steps, whereas T3 retains 89–91% of its peak return across all runs. T3 thus combines consistent return retention across runs with the strongest final goal-reaching score in the evaluated run. Together, these results demonstrate the potential of spatial tactile information while identifying sustained performance as a challenge for some tactile configurations.

3) Common Patterns Across Learning Paradigms: The IL and RL studies reveal two common patterns: compact force-based inputs can improve measured task performance, while richer tactile representations do not consistently provide further gains. T2 exceeds the tactile-free baseline in the reported mean task metric on all three IL tasks and the three single-stage RL tasks. The deformation map achieves strong results in tool use task, but these gains are not stable and consistent across the benchmark. These observations support further investigation into how richer tactile information is encoded and integrated into robotic policies. Together, the studies demonstrate how DextraVerse supports the identification of common patterns in tactile representations and their limits across learning paradigms under study-specific protocols.

B. Component Integration

We demonstrate how DextraVerse supports component integration through two complementary studies. First, simulation-trained RL hand skills assist real-world teleoperation and demonstration collection for downstream IL on the cube-to-basket task (Fig. 3(d)). Second, tactile information is incorporated into RL rewards to guide finger participation and grasp configuration. These studies illustrate how researchers can combine sensing, control, and learning components to develop new manipulation workflows.

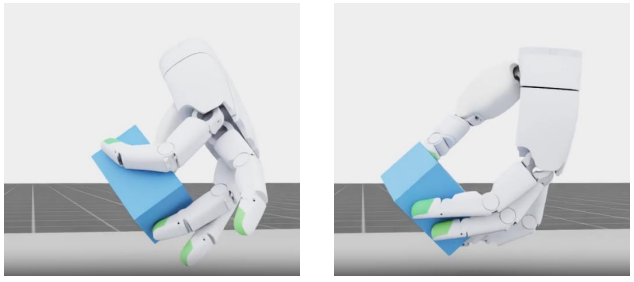
TABLE V: Qualitative comparison of teleoperation configurations by cost, collection efficiency (Eff.), controllability (Con.), and user experience (Exp.), ranked from 1 (best) to 4 (worst).

Wrist	Hand	Cost ↓	Eff. ↓	Con. ↓	Exp. ↓
Tracker	Glove	4	3	**1**	3
Tracker	VR	3	4	2	4
Tracker	RL skills	2	2	4	2
SpaceMouse	RL skills	**1**	**1**	3	**1**

1) RL-Assisted Imitation Learning: DextraVerse connects teleoperation, simulation-trained RL hand skills, and IL through shared control and data interfaces. Independent wrist and hand commands allow researchers to replace direct finger retargeting with an RL controller while reusing the demonstration-recording and downstream learning pipelines. Wrist control uses a Vive Tracker or SpaceMouse; hand control uses VR-based tracking, a glove, or pretrained RL skills. This separation enables learned controllers and operator hardware to be combined within a common collection workflow. Table V summarizes our qualitative engineering assessment of these configurations. Tracker-based wrist control follows natural human motion, whereas SpaceMouse supports deliberate Cartesian adjustments of approximately 5 mm. Direct retargeting preserves individual finger control, while RL skills automate learned hand behaviors without requiring hand-tracking hardware. Operator-side hardware costs approximately $11k for the glove–tracker configuration and $200 for the SpaceMouse–skill configuration, excluding offline policy-training compute.

We demonstrate this integration on the real-world cube-to-basket task. The operator uses SpaceMouse to control the wrist and hands-free voice commands to trigger RL grasping and releasing. After several practice sessions, a single operator collects approximately 75 demonstrations per hour. Mean demonstration duration is 24.8 s, with a complete collection cycle of approximately 47 s including scene re-staging and automatic homing. Of 264 episodes, 261 are retained for training (98.9%). These measurements characterize one operator and configuration; relative collection efficiency requires a matched comparison.

The retained demonstrations feed into the IL pipeline to train a DP with force magnitude and direction as tactile



(a) Baseline (b) With Tactile Reward

Fig. 5: Representative grasps without (a) and with (b) the tactile reward, illustrating increased finger participation around the object.

TABLE VI: Simulation results at final checkpoints (training seed 42). Grasp diagnostics are conditioned on successful episodes.

Metric	Baseline	+ Full-cage
Episode success	204/256 (79.7%)	216/256 (**84.4%**)
All-four ever-contact	75/204 (36.8%)	204/216 (**94.4%**)
Pinky flexion offset	+0.525	**−0.018**
Onset spread (steps)	13.0	**10.5**

inputs. The policy succeeds in 5 of 20 trials (25%), split equally between randomized cube placements on the left and right. Despite limited deployment reliability, this workflow demonstrates how DextraVerse integrates RL skills, teleoperation, and IL through shared interfaces.

2) Grasp-Configuration Reward Design: ***Setup and Metrics:*** To examine how tactile-informed reward design shapes grasp configuration, we add a five-finger contact reward to a baseline with contact-related shaping. Computed from simulated tactile contact signals, this reward encourages simultaneous object contact by the thumb and all four non-thumb fingers. Both policies follow the PPO training protocol for *Cube grasping*, each with a budget of 640 million agent steps. Final checkpoints are evaluated over 256 episodes within a 0–90° wrist-orientation cone, with fixed controller, center-of-mass, and mass randomization settings. Episode success requires at least 10 held steps. For successful episodes, we report all-four ever-contact, mean signed pinky-flexion offset, and median contact-onset spread. All-four ever-contact requires each non-thumb finger to register a contact force above 0.05 N at least once during the episode; simultaneous contact is not required. Pinky-flexion offset is the difference between pinky flexion and the median flexion of the four non-thumb fingers at lift onset, with each finger's flexion computed by averaging its curl-joint positions after normalization to $[-1, 1]$ using joint limits. Contact-onset spread is the within-episode standard deviation of first-contact times among contacting fingers, measured in steps.

Results and Analysis: In the representative grasps shown in Fig. 5, two fingers remain visibly separated from the cube in the baseline, whereas the policy trained with the five-finger contact reward forms a more enclosing grasp. Consistent with this observation, all-four ever-contact increases from 36.8% to 94.4% (Table VI). The mean signed pinky-flexion offset approaches zero, while median contact-onset spread decreases, indicating more closely timed initial contacts among contacting fingers. Observed episode success also increases by 4.7 percentage points. This single-seed comparison provides initial evidence that incorporating tactile contact information into RL rewards can guide finger participation and hand posture alongside task success.

TABLE VII: Comparison of IL algorithms across three tasks.

	Task success (%) ↑		
Task	**DP** [1]	**FM** [23]	**ACT** [2]
Pick-and-place	**67.8** ± 13.9	36.7 ± 3.3	42.2 ± 3.9
Handover	**17.8** ± 7.7	16.7 ± 15.3	**17.8** ± 8.4
Cube stacking	2.2 ± 1.9	1.1 ± 1.9	**4.4** ± 1.9
Task mean	**29.3** ± 5.3	18.2 ± 5.0	21.5 ± 3.2

C. Comparison of IL Algorithms

We next examine how IL algorithm performance varies across tasks. Table VII compares three algorithms on the shared simulated task suite, reporting mean success ± standard deviation across three seeds. Diffusion Policy achieves the highest mean success on *Pick-and-place*, while ACT performs best on *Cube stacking*; the two tie on *Handover*. These task-dependent rankings complement the IL tactile ablation, showing how the shared task suite supports evaluation of both sensory configurations and learning algorithms.

D. AI-Coding Friendliness and Retrieval Efficiency

To assess DextraVerse's support for AI-assisted development, we evaluate whether Repository Retrieval Maps (RRMs) reduce the effort required for coding agents to locate relevant interfaces and trace cross-module dependencies.

1) Setup and Metrics: We compare identical source code with and without RRMs, matching prompts, permissions, timeouts, and execution limits within each model–task pair. Inspection is read-only and excludes skills, subagents, persistent memory, and write tools. Claude Code and Codex receive identical RRM content through their respective repository-instruction interfaces. Seven models complete eight tasks covering entry-point discovery, code localization, summarization of module relationships, and cross-module tracing. Three repetitions per task and condition yield 336 runs. Both conditions use identical scoring criteria based on required code identifiers and answer patterns. For the three integration tasks, no individual RRM contains all the information required to satisfy these criteria.

We measure success, inspection calls (*Calls*), token usage, and wall-clock time. Each inspection-tool or read-only shell invocation counts as one call, regardless of the number of files accessed. Token usage sums reported input and output tokens across model calls, including cached input without double-counting. Mean inspection counts, token usage, and execution times include unsuccessful and audit-flagged runs. Relative reductions are calculated from unrounded means against each model's no-RRM condition and averaged equally across models; success counts are pooled separately.

2) Results and Analysis: RRMs reduce mean inspection calls, token usage, and wall-clock time for all seven models in the full sample, with average relative reductions of 51.3%, 25.2%, and 35.4%, respectively (Table VIII). Pooled

TABLE VIII: Full-sample retrieval performance with and without RRMs across seven coding models.

Model	Successful runs	Reduction (%) ↑		
	Without → with	Calls	Tokens	Time
Sonnet 5	$18/24 \rightarrow 23/24$	54.7	**43.4**	**44.9**
Opus 5	$22/24 \rightarrow 24/24$	55.0	26.3	40.8
Fable 5	$20/24 \rightarrow 23/24$	51.1	19.2	32.2
GPT-5.6 Luna	$19/24 \rightarrow 21/24$	54.5	24.3	29.0
GPT-5.6 Terra	$20/24 \rightarrow 22/24$	**55.7**	32.3	37.2
GPT-5.6 Sol	$24/24 \rightarrow 24/24$	50.5	29.6	35.1
GPT-6 Astra	$23/24 \rightarrow 24/24$	37.3	1.4	28.6
Summary	$146/168 \rightarrow 161/168$	51.3	25.2	35.4

success also increases under the predefined scoring criteria. Beyond retrieval efficiency, RRMs improve adherence to the permitted inspection scope in the evaluated runs. Across the four GPT models, the audit records no scope violations with RRMs (0/96 runs), compared with 16/96 without RRMs. These violations involve searching prohibited content or listing files outside the assigned project directory. Together, the resource savings and observed adherence to inspection boundaries support RRMs as an effective aid for navigating DextraVerse's integrated codebase.

V. Discussion and Challenges

DextraVerse provides reusable infrastructure for investigating tactile-rich dexterous manipulation. Shared control, data, and policy interfaces support tactile representation studies across IL and RL and the composition of learning methods across simulation, demonstration collection, and deployment. The tactile studies identify task-dependent representation benefits and differences between checkpoint performance and long-horizon RL training stability. RL-assisted teleoperation demonstrates the initial feasibility of connecting simulation-trained hand skills to demonstration collection and downstream IL, while RRMs improve repository-navigation efficiency. These studies illustrate how the platform supports both scientific investigation and the development of integrated manipulation workflows. Guided by *Harmony in Diversity, Innovation via Integration*, future research can build on DextraVerse to investigate task-aware tactile fusion, broader skill composition, and reliable bimanual deployment.

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